

Aditya Kulkarni

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SUMMARY

AI engineer with production experience integrating LLMs, RAG architectures, and MCP tool handlers into full-stack applications. Built multi-model LLM routing across Claude, GPT-4o, and Gemini (8+ models) via MCP tooling at Isomer AI (394 commits, 7 deployments). Built a vendor-neutral LLM eval harness tracking hallucination rate, latency, and cost-per-call; a compliance RAG system (Glance AI); and event-driven AI orchestration at 2,000 req/sec (AlgoPulse). TypeScript/React/Node.js full-stack. MS CS at NC State (May 2026). IEEE-published.

EXPERIENCE

- Product Engineer Intern

Isomer AI

May 2025 - Present

 - Architected multi-model LLM routing across Claude, GPT-4o, and Gemini (8+ models) via MCP tool handlers for an AI-powered claims automation platform; built prompt architectures, system instruction sets, and Temporal.io durable workflow orchestration across 7 production deployments (394 commits).
 - Shipped 10+ full-stack features in TypeScript, Node.js, Express, React, PostgreSQL, Redis, and Docker; built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage), RBAC, Auth0 authentication, and CI/CD via GitHub Actions with Sentry observability.
- Graduate Research Assistant

NC State University

Apr 2025 - Present

 - Designed and deployed a cloud-native genomic data platform on Google Cloud Run serving 500GB of datasets to 100+ researchers; improved ML pipeline accuracy by 12% via hyperparameter tuning and time-series analysis.
- Software Developer Intern

Mphasis (HP Company)

Jun 2024 - Aug 2024

 - Built AI-driven automation workflows including a health claim processing tool using OCR, NLP, and ML classification models improving accuracy by 25%; automated CI/CD on AWS and GCP, integrated Grafana monitoring, and supported model validation and red-teaming.
- AI/ML Intern

Capgemini

Jun 2023 - Aug 2023

 - Built distributed data pipelines using Hadoop and Apache Spark for large-scale batch processing and Apache Kafka streaming for real-time ingestion, reducing processing time by 15% and end-to-end latency by 20%.

PROJECTS

- LLM Eval Harness

Responsible AI / LLMOps

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro on 2,000 frozen examples; tracked hallucination rate, factual consistency, p50/p95 latency, and cost-per-call; paired bootstrap significance testing; DuckDB + Parquet columnar storage; Streamlit dashboard for regression detection and automated eval pipelines.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker
- AlgoPulse -- Agentic AI Orchestration

Agentic / MCP

Re-architected from API polling to event-driven streaming after diagnosing throughput bottleneck; Temporal.io durable workflow state with fault-tolerant retries, activity checkpointing, and saga-pattern rollback; Apache Kafka for partitioned worker coordination. Reached 2,000 async LLM inference req/sec with zero message loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka
- Glance AI -- Compliance RAG System

RAG / Responsible AI

Privacy-first RAG system for fintech and healthcare with per-tenant data isolation at the vector retrieval layer; prevents cross-tenant context leakage and LLM hallucinations in regulated environments. Deterministic output architecture with grounded citations enforces responsible AI principles for compliance-sensitive workflows.

Python, LLMs, Vector Databases, RAG, Privacy-First Architecture
- VectorLift -- Semantic Search Engine

NLP / Search

Neural IR pipeline on MS MARCO: BM25 sparse baseline, dense bi-encoder retrieval (sentence-transformers, in-batch negatives), cross-encoder re-ranker; Kafka-driven indexing into Elasticsearch + Qdrant; FastAPI serving with per-stage latency tracing; NDCG@10/MRR@10 evaluation across 6 pipeline configurations.

Python, PyTorch, Elasticsearch, Qdrant, Kafka, FastAPI, Prometheus, Grafana, Docker

EDUCATION

- North Carolina State University -- MS Computer Science

Aug 2024 - May 2026

Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms
- Mumbai University -- BTech Computer Science | CGPA: 9.32/10

Aug 2020 - May 2024

Machine Learning, Big Data Infrastructure, DS & Algorithms, Blockchain

SKILLS & AWARDS

Languages: TypeScript, Python, JavaScript, Java, SQL, C | **AI/ML:** LLMs, MCP, RAG, LangChain, PyTorch, Hugging Face, NLP, OpenAI/Anthropic/Google SDKs | **Frontend:** React, Next.js, Tailwind CSS | **Backend:** Node.js, Express, FastAPI, REST APIs | **Cloud / Infra:** AWS, GCP, Docker, Kubernetes, Kafka, Temporal.io, CI/CD, Grafana | **Databases:** PostgreSQL, MongoDB, Redis, Elasticsearch, Qdrant

HackNC24 Top 10 (100+ teams, 48h) | **Sanjeevani Top 1%** (12 of 1,100+) | **IEEE Published** -- Mapify (500+ attendees) | **Google DSC Lead** (500+ students)